Gears, motor and first power

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Learning target: Compare a gear-based prediction with a supervised first powered test.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Ideal speed becomes 3 times as large. Actual rpm, load or slipping may change.

02. Worked example: explain the model
One rotation 0.20 m; 2 rotations/s gives 0.40 m/s.

03. Read and interpret the model
Actual labeled configuration with inspection observations.

04. Apply the physics
0.30 m/s.

05. Explain your reasoning
Loaded rpm, slip, gear rubbing, wheel circumference; measure loaded wheel rpm or marked-wheel rotations over distance.

06. Review: connect an earlier idea *Review: Unit 7, Day 12.*
A stronger field at A in that drawing convention; field lines are a model, not physical strands.

07. Exit check
Actual checkpoint and evidence; leave approval unmarked until granted.

08. STRETCH: test the model
Slip, wrong effective circumference, timing/rotation measurement or changing speed. Track wheel and ground motion together.

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01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Torque ratio 5; speed ratio 1/5. Assumes ideal gears with negligible losses.

02. Worked example: explain the model
One rotation 0.20 m; 2 rotations/s gives 0.40 m/s.

03. Read and interpret the model
Actual labeled configuration with inspection observations.

04. Apply the physics
160 wheel rpm; 0.4189 m/s; 7.16 s.

05. Explain your reasoning
Loaded rpm, slip, gear rubbing, wheel circumference; measure loaded wheel rpm or marked-wheel rotations over distance.

06. Review: connect an earlier idea *Review: Unit 7, Day 12.*
Each piece is normally a smaller magnet with both poles; cutting does not isolate a magnetic monopole.

07. Exit check
25% reduction; drivetrain load lowers actual rotational speed.

08. STRETCH: test the model
Ideal $2\pi \times 0.025 \times 4 \approx 0.628$ m/s; ratio≈ 0.796. Other measurement/model errors also possible.